MACHINE LEARNING ENGINEER · Open to jobs & internships

Mustofa A. Ahmed

I make large models run fast on real hardware: LLM inference, speculative decoding and model deployment across GPUs, FPGAs and NPUs. PhD student at UCAS / IHEP, Chinese Academy of Sciences.

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Experience

Algorithm R&D Intern · Zidong Taichu (紫东太初) — Jul – Oct 2026 · Beijing

- Reproduced DFlash and DSpark speculative decoding for Qwen3.5-4B; built a DSpark-capable vLLM runtime.
- Fixed model-loading, hybrid-attention and CUDA Graph issues; benchmarked on GSM8K, HumanEval and MT-Bench (RTX 4090).

Research Fellow · Computing Center, IHEP, CAS — 2023 – present · Beijing

- ML deployment for particle-physics experiments CEPC and CMS (CERN): ParticleNet / ParT on TensorRT and ONNX.
- Built AutoDeploy and its deployment knowledge graph; trained DFlash drafters for vLLM serving studies.

Web Developer · GeePower Energy Technology — 2021 – 2022 · Yangzhou

- Full-stack company website; database design and query optimization.

Selected projects



AutoDeploy
Model-to-deployment platform: inspects a model, picks an optimization path (TensorRT, ONNX, quantization, pruning) and deploys to GPU, FPGA or NPU.
10.4× ParT · 6.6× ParticleNet



Speculative decoding
DFlash, DSpark and MTP speculative decoding in vLLM; trained a 3-layer DFlash drafter and studied proposal-budget trade-offs.
up to 3.03× faster decoding



FPGA & GPU acceleration
Deployed networks on Xilinx FPGAs (incl. Alveo) via HLS, hls4ml and Vitis AI; tuned for NVIDIA GPUs and Ascend NPUs.
GPU · FPGA · NPU



Knowledge graphs & GNNs
Neo4j knowledge graph linking models, techniques, tools and hardware behind AutoDeploy; graph learning with HGT and TGN.
Neo4j · HGT · TGN

Qwen3.8

Personal 24/7 AI assistant
Always-on assistant built on Qwen3.8-27B, served locally with llama.cpp and driven by Hermes Agent for tool use.
Personal project · ongoing



Unreal Engine 5 & MetaHuman
World-building, interactive scenes and digital-human design; plus AI-generated music videos.
Creative side

Publications

2026 **Draft-budget trade-offs in block-parallel speculative decoding: an empirical study.** Mustofa, Cheng. Submitted to IEEE CSCWD 2027

2026 **A knowledge-driven framework for HEP model deployment on heterogeneous hardware.** Mustofa, Cheng. *Radiation Detection Technology and Methods*. doi ↗

2026 **MCG: A multi-data-center content gateway for topology-hiding and secure WAN storage access.** Wei et al. *29th IEEE CSCWD 2026*, 2738–2743. doi ↗

2025 **Enhancing GRB detection: Bayesian optimization of triggerless data analysis for LHAASO-WCDA.** Mustofa, Cheng, Yao. *EPJ Web of Conferences* 337. doi ↗

2024 **Fast lossless image compression for synchrotron radiation facility using deep learning.** Minxing et al. *Radiation Detection Technology and Methods* 8(4). doi ↗

2023 **SECHA: A smart energy-efficient, cost-effective home automation system for developing countries.** Mustofa et al. *J. Computer Networks and Communications*. doi ↗

Education

PhD, Computer Application Technology
UCAS / IHEP, CAS · 2026 – present

M.Eng., Computer Application Technology
UCAS / IHEP, CAS · 2023 – 2026

B.Sc., Microelectronics Engineering
Yangzhou University · 2019 – 2023

Skills

LLM INFERENCE & SERVING
vLLM, llama.cpp, DFlash, DSpark, MTP, CUDA Graphs

MODEL OPTIMIZATION
TensorRT, ONNX, quantization (PTQ, QAT), pruning, hls4ml, Vitis AI

DEEP LEARNING
PyTorch, Hugging Face, GNNs (HGT, TGN), RL

PROGRAMMING & INFRA
Python, C++, CUDA, Linux, Docker, Git, CI/CD, Neo4j

HARDWARE
NVIDIA GPUs, Xilinx FPGAs, Huawei Ascend NPUs

LANGUAGES
English, Amharic, Mandarin (HSK 4)

Awards

2026 CSC Chinese Government Scholarship
2026 IHEP Director's Award, Best Graduation Project
2023 ANSO Scholarship for Young Talents
2021 Yangjie "Excellent Student" Scholarship

Research interests

ML deployment & acceleration · Efficient LLM inference · RL · World models · Autonomous agents

Hobbies

Volleyball, badminton, swimming · Spurs fan

Add me on WeChat
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